Light's Criteria

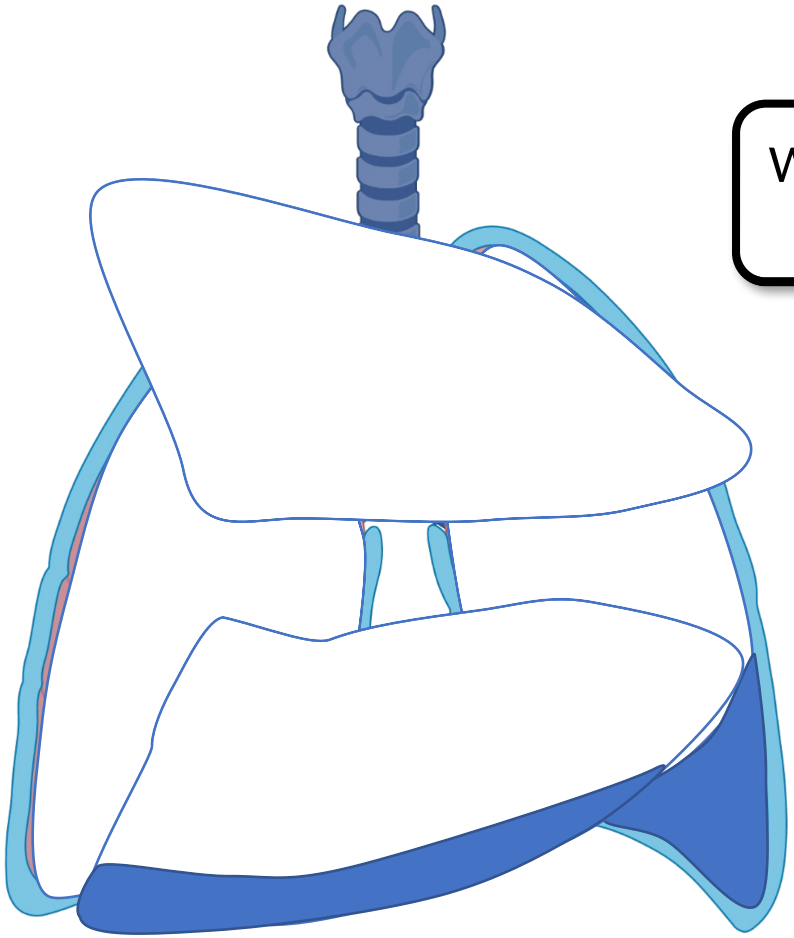
Differential

Management

Cases

Click these buttons to **skip to each learning objective**

Pleural Effusions



Which pleural effusions warrant further work-up?

Nearly all pleural effusions > 1 cm on US or lateral decubitus CXR should be further evaluated with a thoracentesis.

Image from biorender.com

teachIM.org

Light’s Criteria
Differential
Management
Cases

LIGHT’S CRITERIA

Pleural: serum protein >0.5
Pleural: serum LDH >0.6
Pleural LDH >2/3 ULN of serum LDH

0 criteria → TRANSUDATE

≥1 → EXUDATE

Pleural studies

- Protein
- LDH
- Gram stain and culture
- Cell count
- pH
- Glucose

Specialized tests

- Cytology/ flow cytometry
- Adenosine deaminase
- Amylase
- Triglycerides
- Hematocrit

Click on “Bonus information” to learn more about pleural fluid testing.

Bonus information

Light's Criteria

Differential

Management

Cases

LIGHT'S CRITERIA

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TRANSUDATE

Heart failure
Cirrhosis (Hepatic hydrothorax)
Nephrotic syndrome
Hypoalbuminemia

Peritoneal dialysis
Atelectasis
Trapped lung

EXUDATE

Exudate	Fluid studies	Specialized tests
Infection (bacterial, TB)	Bacterial: ↑ PMNs	
	TB: ↑ lymphs	↑ ADA, ± AFB stain/culture
Parapneumonic	↑ PMN	
Malignant	↑ lymphs	+ cytology/ flow
Hemothorax	↑ RBCs	Pleural hct:serum hct > 0.5
Chylothorax	↑ lymphs	TG > 110
PE	Bloody effusion	
Autoimmune (RA, SLE, GPA, Sjogren's, sarcoidosis)	↑ lymphs	
GI (esophageal perf, pancreatitis)	Perf: ↑ PMNs	↑ amylase
	Panc: ↑ lymphs	

Pleural studies

Protein
LDH
Gram stain and culture
Cell count
pH
Glucose

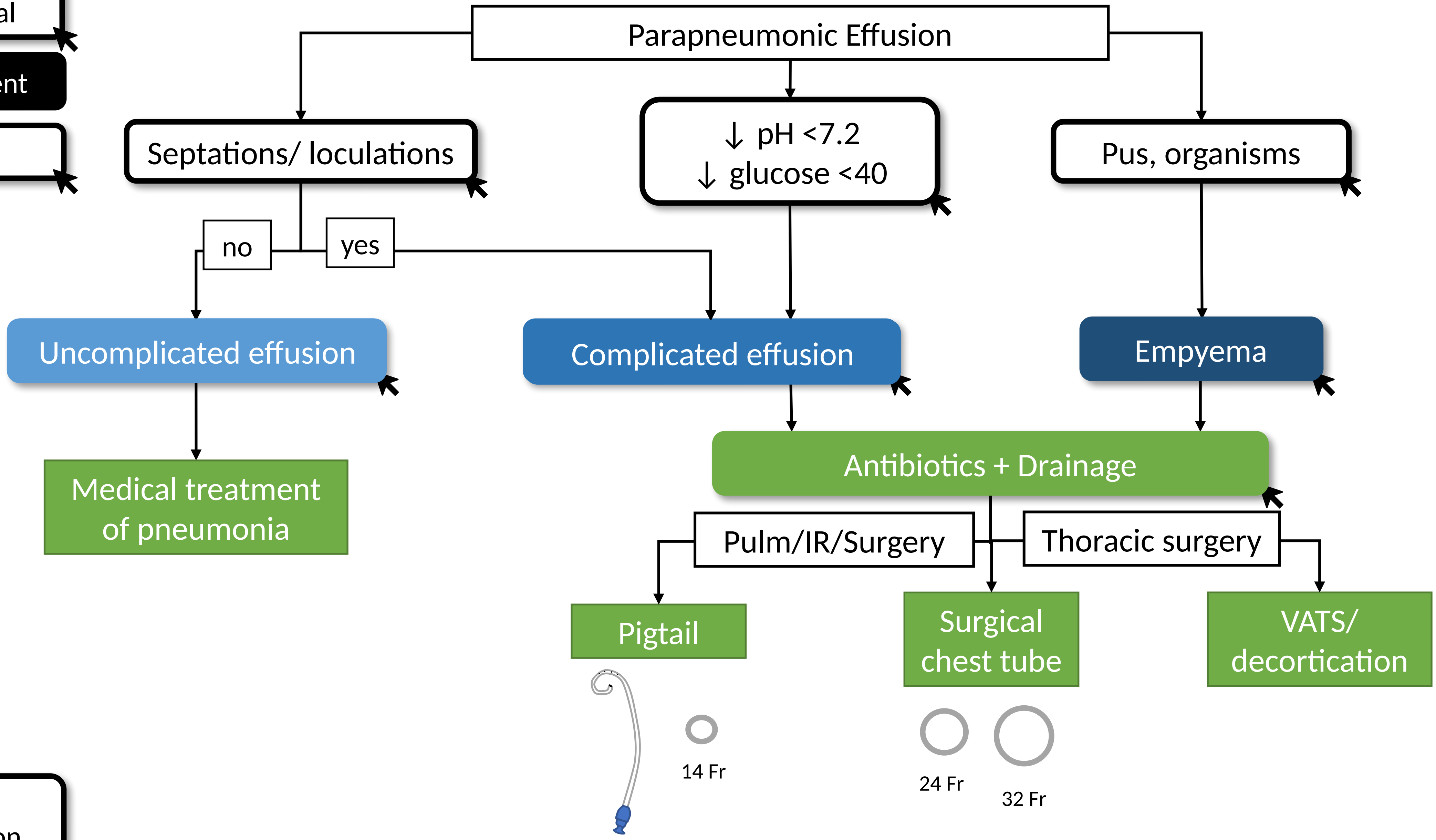
Specialized tests

Cytology/ flow cytometry
Adenosine deaminase
Amylase
Triglycerides
Hematocrit

Fluid studies

Bonus information

Management of Parapneumonic Effusions



Bonus information

Light's Criteria

Differential

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Cases

Case 1

46 yo presents with fevers, cough, and SOB. CXR shows a L sided consolidation and moderate pleural effusion.

<u>Pleural fluid studies</u>

Protein 3.5
LDH 76
pH 7.26
Gram stain: + PMNs
Cell count: 350 (88% PMNs)

Serum protein: 4.0
Serum LDH: 278

Exudate. Meets > 1 Light's criteria (pleural: serum protein > 0.5)

Not needed.

Uncomplicated parapneumonic effusion

Treatment of pneumonia

Case 2

72 yo with a prior CVA and chronic aspiration presents with generalized weakness, chest pain and cough. CT scan shows a loculated R sided effusion with septations and RLL consolidation.

<u>Pleural fluid studies</u>

Protein 3.5
LDH 356
pH 7.16
Gram stain: + PMNs
Cell count: 980 (90% PMNs)

Serum protein: 3.5
Serum LDH: 278

Exudate. Meets all 3 Light's criteria.

Not needed.

Complicated parapneumonic effusion

Antibiotics + drainage of the loculated effusion (pigtail vs. chest tube). tPA may be used to help drainage.

Case 3

82 yo with metastatic breast cancer presents with new onset SOB. CXR shows near opacification of the L hemithorax. Therapeutic thoracentesis is done.

<u>Pleural fluid studies</u>

Protein 2.1
LDH 156
pH 7.46
Gram stain: + none
Cell count: 580 (80% lymphocytes)

Serum protein: 2.5
Serum LDH: 278

Exudate. Meets > 1 light's criteria (↑ pleural: serum protein and LDH ratios).

Cytology: negative
Teaching point: In effusions of uncertain etiology, send cytology as part of the initial work-up (British Thoracic Society guidelines)

Likely malignant effusion

Repeat thoracentesis.

Repeat thoracentesis. When looking for a malignant effusion, sending serial thoracenteses increases the sensitivity.

Bonus information

Light's Criteria

Differential

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Cases

Pleural studies

Protein, LDH

Gram stain and culture

Cell count

pH

Glucose

Specialized tests

Cytology
Flow cytometry

Adenosine deaminase

Amylase

Triglycerides

Hematocrit

Nucleated cells: The number of nucleated cells can help narrow the differential.

>50k — Complicated parapneumonic, empyema

10-50k — Uncomplicated parapneumonic, Acute pancreatitis, SLE pleuritis

The utility of pleural fluid pH is controversial because of several factors that can influence its accuracy (exposure to air or

A low pleural glucose <60 suggests high metabolic activity and is seen in infections, malignancy, and autoimmune diseases.

- TB pleurisy

British Thoracic Society guidelines recommend sending cytology in pleural effusions of unknown etiologies. Flow cytometry should be sent only in patients with suspected leukemia/lymphoma.

Serial pleural cytologies ↑ sensitivity for malignant effusions.

Cytology	Sensitivity
1	65%
2	92%
3	97%

The presence of organisms on gram stain or a positive culture is diagnostic of an empyema.

An ADA > 40 is highly sensitivity and specific for TB pleurisy (>90%). This is more sensitive than sputum AFB, pleural fluid AFB,

Typically seen in esophageal rupture, pancreatitis, or occasionally, malignancy.

Triglyceride levels > 110 mg/dL is suggestive of a chylothorax.

Pleural fluid to blood hematocrit ratio > 0.5 is suggestive of hemothorax.

Pleural Effusions — 2023 Context Update

Light's criteria, the transudate/exudate differential, and the parapneumonic framework are timeless and verified current (June 2026). The items below add modern context, not corrections.

Light's criteria unchanged: exudate if ANY one of pleural/serum protein >0.5, pleural/serum LDH >0.6, or pleural LDH >2/3 serum ULN (sens ~98%, spec ~83%). In diuretic-treated patients it misclassifies ~25% of transudates — use serum–pleural albumin gradient >1.2 g/dL to reclassify.

BTS 2023 pleural disease guideline: pleural fluid pH ≤7.2 (or frank pus / positive Gram stain/culture) indicates complicated effusion/empyema → drain. Small-bore (≤14F) chest tube is first-line for pleural infection (less pain, equal efficacy); larger bore reserved for heavy loculation or pus.

Intrapleural tPA + DNase (MIST-2, Rahman NEJM 2011) improves drainage and reduces surgical referral but does NOT lower mortality — the deck's caveat still holds. RAPID score risk-stratifies pleural infection prognosis.

Malignant effusion: send cytology on initial work-up; serial large-volume thoracenteses raise yield (~65% → ~92% on a 2nd). For recurrent symptomatic MPE, indwelling pleural catheter and talc pleurodesis are both first-line (ATS/STS/STR 2018); IPC + talc (IPC-PLUS) increases pleurodesis success.

Take-Home Points

1. Evaluate any pleural effusion >1 cm on US/lateral decubitus with an ultrasound-guided diagnostic thoracentesis.

2. Light's criteria separate transudate (non-inflammatory: heart/renal/liver failure, hypoalbuminemia) from exudate (inflammatory: infection, malignancy, autoimmune, PE, chylothorax). Any one criterion = exudate.

3. Pleural fluid studies localize the cause: cell count/differential (PMN vs lymphocyte), pH/glucose, Gram stain/culture, cytology, ADA, amylase, triglycerides, hematocrit.

4. Parapneumonic effusions: uncomplicated → antibiotics alone; complicated (pH <7.2, low glucose, loculations) and empyema (pus or +culture) → antibiotics PLUS drainage, often multidisciplinary (pulm/IR/thoracic surgery).

5. tPA/DNase augments drainage without a mortality benefit; small-bore chest tubes are first-line for pleural infection (BTS 2023).

Attribution & Optimization Notes

Original lesson: “Pleural Effusions,” TeachIM.org (updated Apr 2022).

Authors: Neha Deshpande MD & Ryan Murphy MD; Executive Editor Yilin Zhang MD.

Source URL: teachim.org/teaching_material/pleural-effusions/

Edit type: APPEND-ONLY. No surgical text change made — the standing teaching is accurate and contains no spelling/number errors. All 6 original click-to-reveal animated slides and BioRender graphics preserved.

Human review: Case 3 (slide 5, malignant effusion) labels the exudate as meeting “protein and LDH ratios,” but the LDH ratio is 156/278 = 0.56 (below the 0.6 threshold). It clearly meets the protein ratio (0.84); whether it meets the absolute LDH criterion depends on the lab’s serum LDH ULN (not given). Diagnosis/management are correct — wording left unchanged pending author confirmation.

Context added (source-cited): BTS 2023 pleural disease guideline; MIST-2 (Rahman NEJM 2011); ATS/STS/STR 2018 malignant effusion. No companion handout is published for this lesson. Medical content should be clinician-reviewed before teaching.